MILITARY SPECIFICATION

CASE, CARTRIDGE, 30MM AMMUNITION FOR M788 M789 and M799 SERIES OF 30MM AMMUNITION

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers, requirements, quality assurance provisions and packaging of the 30MM Ammunition Cartridge Case for use in M788, M789 and M799 series of 30MM Ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-6088	-	Heat Treatment of Aluminum Alloys
MIL-I-6866	-	Inspection, Penetrant Method of
MIL-A-8625	-	Anodic Coating, Aluminum and Aluminum Alloys
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-63547	-	Material Specification for Aluminum Alloy, 7475, Rod

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-115	-	Federal Test Method Standard, Metals; Test Methods

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests
MIL-STD-651	-	Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

11825902	-	Case, 30MM Cartridge
11825947	-	Cartridge, 30MM, M788, TP
11825969	-	Test Barrel, 30MM Mann (6.5°)

PUBLICATIONS

AMCR 715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD, adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

STANDARDS

American Society for Testing and Materials (ASTM)

ASTM E4	-	Verification of Testing Machines
ASTM E8	-	Tension Testing of Metallic Materials
ASTM E18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of the conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

3.1.1 Materials shall be in accordance with the applicable drawings and specifications.

3.1.2 Cartridge case. The cartridge case shall comply with all requirements specified on Drawing (Dwg. 11825902), all associated drawings and with all requirements specified in applicable specifications.

3.2 Characteristics.

3.2.1 Performance. The cartridge case shall meet the following performance requirements when assembled in a 30MM Target Practice Cartridge as shown on Drawing 11825947. Unless otherwise specified, performance requirements involving the firing of the cartridge, apply when fired in a Mann barrel conforming to Drawing 11825969.

3.2.1.1 Case rupture. When fired under the conditions of 3.2.1, the case shall not rupture. It shall also not rupture when assembled in specially loaded cartridges and subjected to an overpressure condition of from 380 to 415 MPa.

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3.2.1.2 Case extraction. Following firing under the conditions of 3.2.1, the case shall be capable of being extracted from the gun chamber without failure of the extractor ring.

3.2.1.3 Projectile extraction. When subjected to an extraction pull at a rate of 25MM per minute ± 10%, the cartridge case shall provide a retention force such that the mean pull value less two times the standard deviation shall be greater than 8 kiloNewtons.

3.2.1.4 Automatic gun operation. The case, as an integral part of the complete cartridge, shall be capable of performing as required in the M230E1 Gun System and Aden MK IV Gun System.

3.2.1.5 Manufacturing process. The cartridge case shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval (see 6.9).

3.2.1.6 Function and casualty. The sample cartridge cases shall be assembled into cartridges and shall function without casualty when tested as specified.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The case shall conform with the dimensions shown on Drawing 11825902.

3.2.2.2 Weight. The weight of the case shall be 56 grams nominal (for information only).

3.2.2.3 Volume. The internal volume of the case shall be 67 cubic centimeters minimum.

3.2.2.4 Specific materials characteristics. The cartridge case shall have the specified mechanical characteristics as listed in Tables I and II. The areas from which sample sections of the case are used for testing are shown in Figures 1 and 2.

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TABLE I. Hardness and mechanical property requirements for average values.

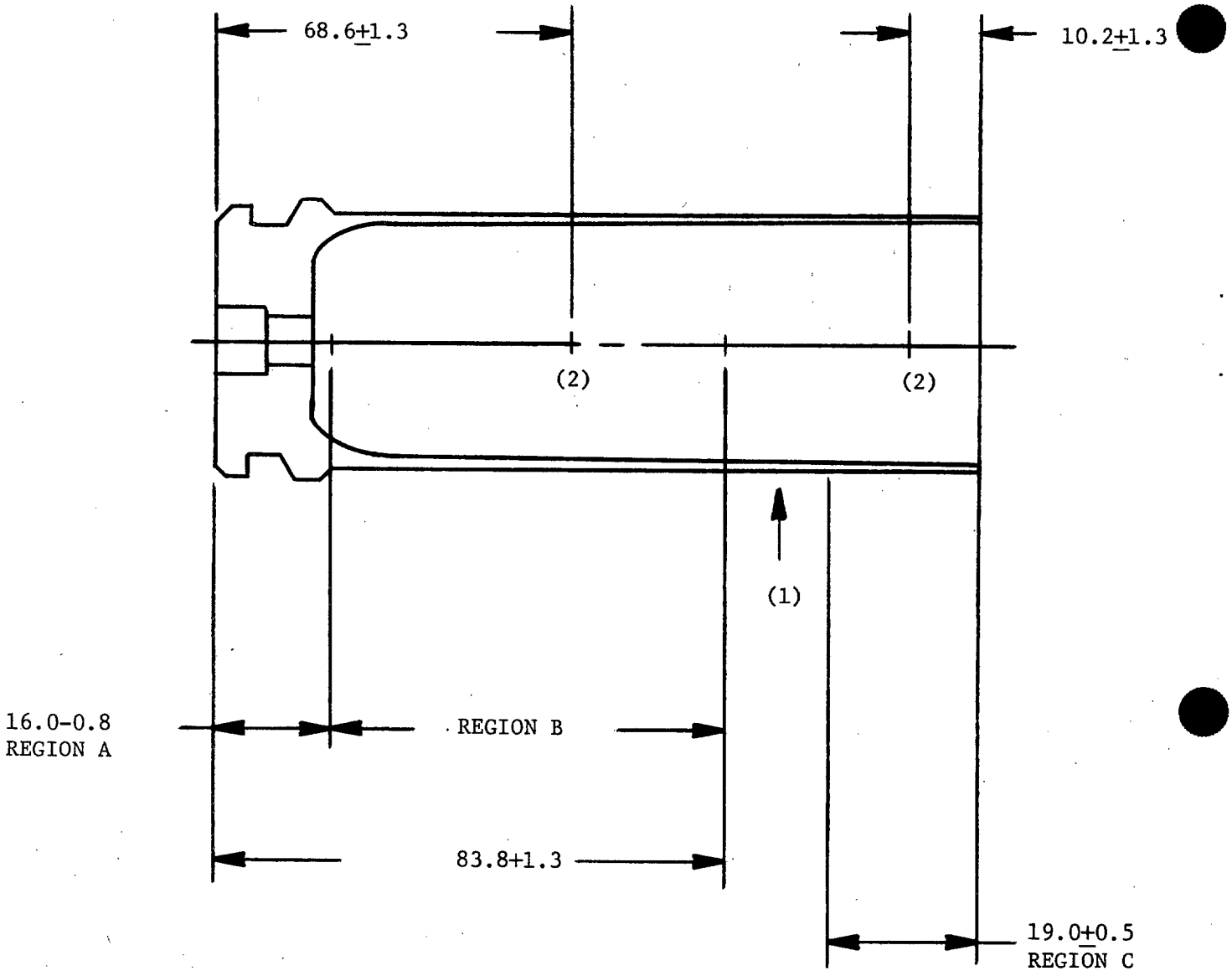
Characteristic	Region A	Region B		Region C	
	Min	min.	max.	min.	max.
Tensile Strength, psi	---	74,000	---	50,000	62,000
Yield Strength, psi	---	57,000	73,000	36,000	44,000
& Elongation, 1 inch	---	7	14	7	16
Hardness Rockwell 30-T 1/	---	68	---	54	61
Hardness Rockwell B	85	---	---	---	---

TABLE II. Hardness and mechanical property requirements for individual coupon values.

Characteristic	Region A	Region B		Region C	
	Min	min.	max.	min.	max.
Tensile Strength, psi	---	73,000	---	49,500	63,000
Yield Strength, psi	---	56,000	73,000	35,000	44,500
& Elongation, 1 inch	---	6	14	6	16
Hardness Rockwell 30-T 1/	---	68	---	53	62
Hardness Rockwell B	85	---	---	---	---

1/ Hardness requirements apply for the inside diameter of a bare case for both Regions B and C. If readings are taken with anodize finish, a correction factor as established in accordance with 4.5.3.10.2 shall be applied.

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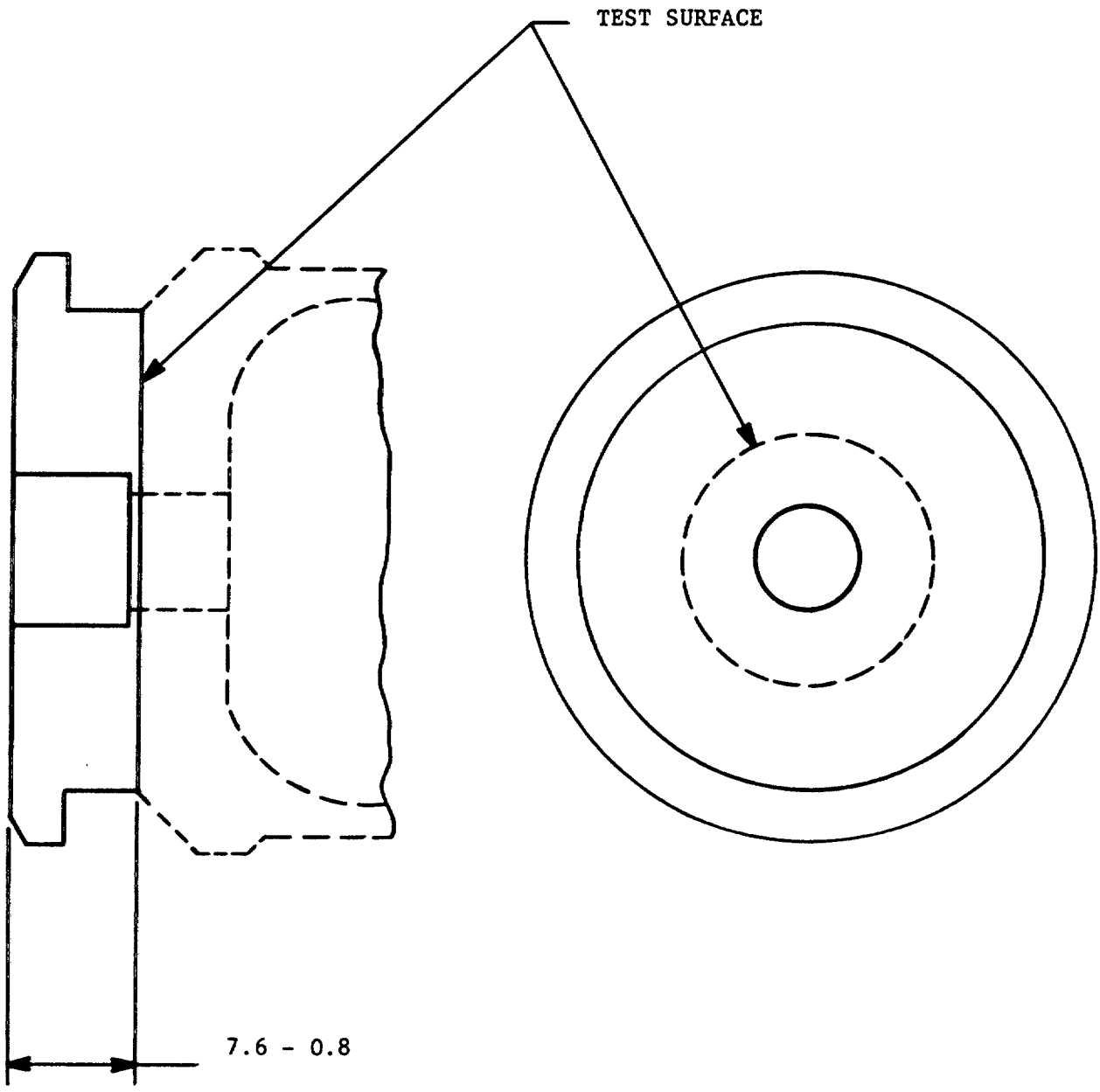
NOTES:

(1) The heat affected area adjacent to the mouth annealed zone will have a hardness gradient spanning the valves found in regions B and C.

(2) Location of centerlines of circumferential tensile speciments. The test coupons from this area shall also be used for the hardness test.

TEST SPECIMEN AREAS
FIGURE 1

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HEAD HARDNESS TEST SAMPLE CONFIGURATION

FIGURE 2

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3.2.3 Environmental conditions.

3.2.3.1 Operating. The case shall not be damaged and shall function as required during or after exposure to the following environmental conditions. Unless otherwise provided the conditions apply with the case assembled into a complete cartridge, or simulation thereof. See Drawing 11825947.

3.2.3.1.1 Temperature. Temperature ranging from -54°C to 71°C.

3.2.3.1.2 Mechanical property stability. Exposure to temperature of 121°C for 24 hours.

3.2.3.1.3 Salt spray. Exposure to salt sea atmosphere. (Ref: MIL-STD-331, Method 107).

3.3 Design and construction.

3.3.1 Configuration. The case shall conform to the requirements specified on Drawing 11825902 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3.2 Standards of manufacture.

3.3.2.1 Anodic coating. The case shall be anodized according to paragraph 7.2.2 of MIL-STD-171 except the coating thickness and coating weight shall be as specified in Figure 3.

3.3.2.2 Material. The case shall be manufactured from aluminum alloy, as specified in specification MIL-A-63547 (AR).

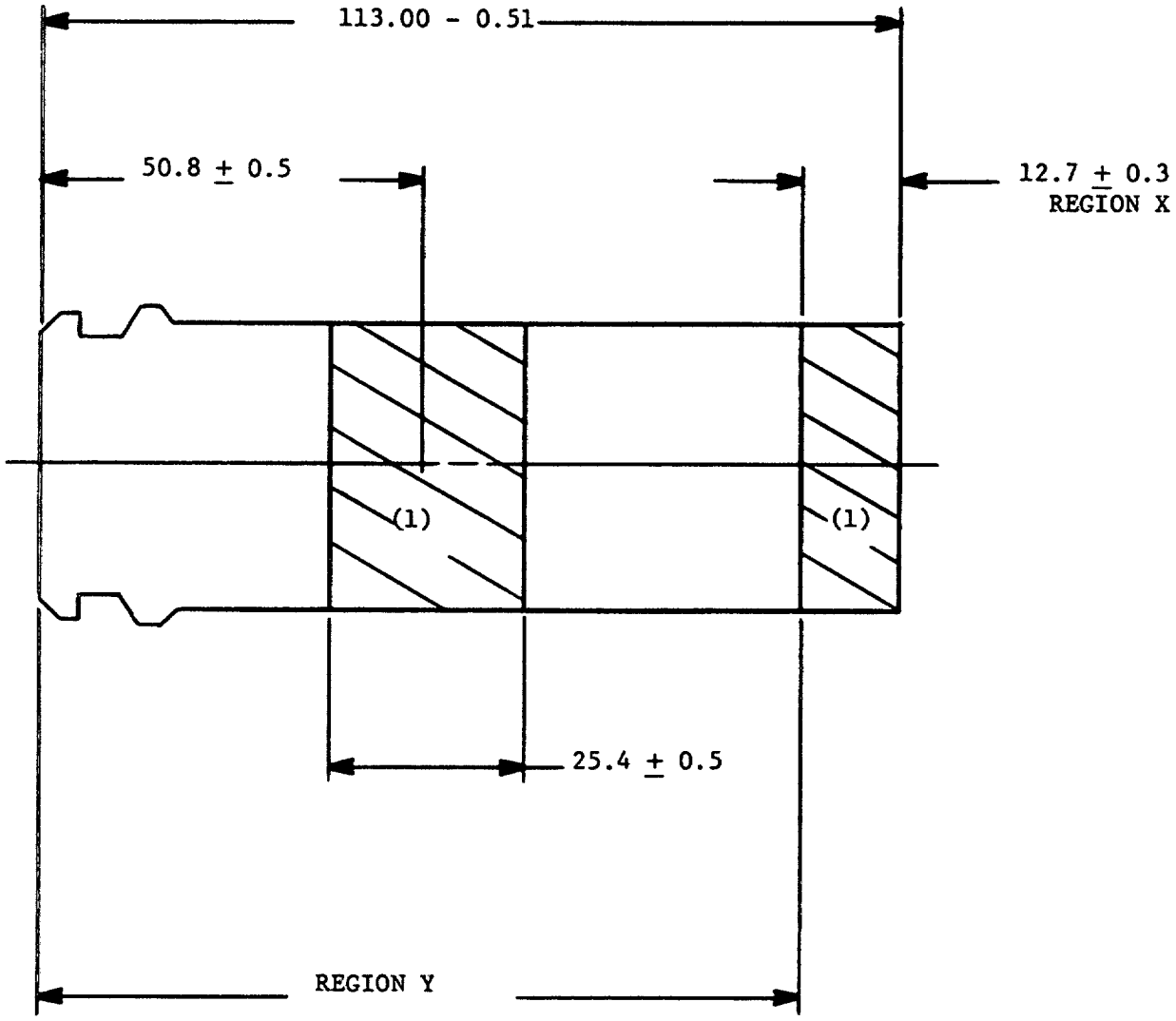
3.3.2.3 Heat treatment. The heat treatment process used in fabrication of the case shall be in accordance with MIL-H-6088.

3.3.2.4 Temper. The case shall be solution treated and aged to the T-76 type condition.

3.3.2.5 Case mouth properties. The case mouth properties, as shown in Region C of Figure 1, shall be obtained by using a separate thermal treatment process. The temperature of the heating medium used for this process shall not exceed 600°F. A time delay of twenty (20) seconds minimum shall be used between the thermal treatment and any water rinse to avoid partial solution of the hardening constituents in the aluminum alloy.

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REGION	MINIMUM WEIGHT	THICKNESS (MILLIMETERS) INSIDE	OUTSIDE
X	2.368 mg/cm^2	0.008 MIN	0.011 to 0.024
Y	2.368 mg/cm^2	CONTINUOUS COATING	0.011 to 0.024



(1) Location and size of samples for coating thickness and weight determinations

COATING WEIGHT AND THICKNESS REQUIREMENTS

FIGURE 3

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During the mouth thermal treatment process a routine hardness test using a method as defined in 4.5.3.4.3, shall be taken to assure continued conformance to the desired parameters and requirements of Table II. At least one cartridge case in each 500 produced shall be tested in this manner. The test results shall be recorded and be available for review by the Government. Prior to start of production the contractor shall submit a copy of the mouth anneal procedure and data verifying the procedure as described in 6.6, to the Government for review.

3.3.2.6 Cleaning methods. Cleaning methods used shall not be detrimental to the case nor shall the cleaning agent contaminate the part or have an adverse effect on subsequent assembly operations.

3.3.2.7 Thermal treatment rework. All cases which are reworked through any thermal treatment must still meet the characteristics of this specification. Any cases requiring this type of rework must be segregated and identified as a separate lot or sublot. The history of the thermal rework shall be recorded and submitted with the other acceptance test data for that lot. The Government shall be notified of the rework characteristics before the lot is shipped and acceptance of the lot shall be dependent upon Government approval of the rework procedure. Whenever mouth anneal rework is performed, additional testing of hardness at the forward most position of Region B shall be done to confirm that upper sidewall properties have not been degraded. This hardness test information shall become part of the acceptance test data submitted with the lot.

3.4 Preproduction sample. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. The case shall be free of defects such as metal flaws, cracks, splits, perforations, buckles, folds, and dents, as well as protective coating and marking deficiencies which exceed the visual defect standards (Ref. 6.4). The case shall be clean and free of foreign matter such as corrosion, dirt, grease, oil, lint, chips, slivers, and shavings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First aritlce inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evalation in accordance with the provisions of 4.3.2.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table III.

4.3.2.1 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract drawings and specification requirements.

4.3.3 Rejection. See MIL-A-48078.

4.3.3.1 First article sample failure. Failure of the sample to comply with and of the requirements of the drawings and specifications shall result in sample disapproval.

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TABLE III. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63976(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Case, cartridge, 30MM		11825902
			NEXT HIGHER ASSEMBLY 11825901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	25		3.3.1	4.4.2.1 (Except majors 108, 109 & 110)
	Hardness	5		3.2.2.4	4.4.3.2
	Salt Spray	20		3.2.3.1.3	4.5.2.3
	Tensile Strength	5		3.2.2.4	4.4.3.1
	High Pressure	30		3.2.1.1	4.5.1.1.1
	Projectile Extraction	20		3.2.1.3	4.5.1.3
	Case Extraction	20		3.2.1.2	4.5.1.2
	Function & Casualty			3.2.1.6	4.5.2
	Ambient (+21°C) ±5°C	60		3.2.1	4.5.2
	High Temperature (+71°C) ±5°C	60		3.2.1	4.5.2.2
	Low Temperature (-54°C) ±5°C	60		3.2.1	4.5.2.1

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63976(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 11825902
4.4.2.1	Case, cartridge 30MM				NEXT HIGHER ASSEMBLY 11825901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined	See 4.4.2a			
Major					
101	Split or perforated case		100%	3.3.1	Visual
102	Profile and alignment, max 2/ (chamber interface)		100%	3.3.1	Gage
103	External defects, sidewall		100%	3.3.1	Test (4.5.3.7.1)
104	External defects, cartridge case mouth		.25%3/	3.3.1	Test (4.5.3.7.2)
105	Internal defects, sidewall		.25%	3.3.1	Test (4.5.3.7.3)
106	Presence of mouth anneal		.25%	3.3.1	Test (4.5.3.10.1)
107	Completeness of heat treatment		.25%	3.3.1	Test (4.5.3.10.2)
108	Hardness	5	Acc-0 Rej-1	3.2.2.4	4.4.3.2 5/
109	Tensile strength	5	Acc-0 Rej-1	3.2.2.4	4.4.3.1
110	Function & casualty	200			
	Ambient Temp	100	0-1*	3.2.1	4.5.2
	Low Temp (-54°C)	50	0-1*	3.2.1	4.5.2.2
	High Temp (+71°C)	50	0-1*	3.2.1	4.5.2.1
111	Projectile extraction	20		3.2.1.3	4.5.1.3

NOTES: * Accept-Reject Criteria

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 11825902
4.4.2.1	Case, cartridge 30MM				NEXT HIGHER ASSEMBLY 11825901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Rim thickness		.40%	3.3.1	Gage
202	Extractor groove diameter, max.		.40%	3.3.1	Gage
203	Overall length		.40%	3.3.1	Gage
204	Location of primer counterbore hole		.40%	3.3.1	Gage
205	Corrosion		.40%	3.3.1	Visual
206	Sidewall basic length min (17.78) 4/		.40%	3.3.1	Gage
207	Sidewall basic length min (83.82) 4/		.40%	3.3.1	Gage
208	Scratch, bend, bulge, wrinkled or fold		.40%	3.3.1	Visual
209	Anodic coating thickness on mouth O.D. (min)		.40%	3.3.1	Test (4.5.3.5)
210	Primer counterbore depth		.40%	3.3.1	Gage
211	Primer counterbore, dia.		.40%	3.3.1	Gage
212	Length to basic shoulder dia., min.		.40%	3.3.1	Gage
213	Flash tube hole dia.		.40%	3.3.1	Gage
214	Foreign matter (except corrosion)		.65%		Visual
215	Protective coating damaged or incomplete		.65%	3.3.1	Visual
216	Marking incorrect, illegible or missing		.65%	3.3.1	Visual
217	Inside diameter of mouth		.65%	3.3.1	Gage
218	Wall thickness at mouth		.65%	3.3.1	Gage
219	Extractor groove width, min		.65%	3.3.1	Gage
220	Location of flash tube hole		.65%	3.3.1	Gage
221	Sidewall thickness (2 places)		.65%	3.3.1	Gage
222	Evidence of poor workmanship		.65%	3.3.3	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63976(AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.1	Case, cartridge 30MM		11825902
			NEXT HIGHER ASSEMBLY 11825901

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ Visual defects shall be evaluated in accordance with visual defect standards. See 6.4. 2/ If a chamber gage is used, a maximum force of 110 Newtons may be used to insert the case into the gage. 3/ Min. sample size 315 no defects allowed. 4/ An undersize diameter is permissible if such undersize condition is caused by ovality or flat spots within visual defect criteria, provided that the total length of the affected area does not exceed 10% of the part circumference. 5/ Failure of two or more sample cartridge cases to comply with the average hardness requirement at one or more locations shall be cause for rejection of the lot. If one cartridge case of the sample fails to comply with the requirements, a second sample consisting of the same number of cartridge cases as the first sample shall be tested. If in the accumulated sample, two or more cartridge cases fail to comply with the average hardness requirements, the lot shall be rejected. 6/ The lot will be rejected if any malfunction or firing casualty of Table IV occurs in number(s) equal to or greater than the applicable rejection (Rej) number. If malfunctions or casualties occur in excess of the applicable acceptance (Acc) number, but less than the "Rej" number, a second sample of double the number of the first will be refired in gun. The lot will be rejected if in the combined samples malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number. The ambient, high and low temperature phases will be considered as separate tests.				

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4.4.3 Testing.

4.4.3.1 Mechanical properties. A sample quantity of five cases shall be subjected to the mechanical property tests of paragraph 4.5.3.4.1. If the average of the data from these five cases for any one property does not meet the allowable ranges of Table I or if an individual data point does not meet the requirements of Table II the lot shall be rejected.

4.4.3.2 Hardness. Hardness tests shall be conducted on the same sample of five cases as used for the mechanical tests of 4.4.3.1 in accordance with the procedures of 4.5.3.4.2. In Region A, if any of the individual hardness values are less than the allowable minimum shown in Table II the lot shall be rejected. In Region B and C in which hardness is taken on tensile coupons, the lot shall not be rejected for hardness if all the mechanical property characteristics are within the specified requirements.

4.4.3.3 In-process inspections or tool controlled dimensions. All characteristics and dimensions not otherwise checked as a requirement of this specification shall fall into this category. This inspection shall include the presence of internal visual defects such as those defined in 6.4. If a defect is found in this category, the supplier will be notified immediately to correct the defect. Cartridge cases completed or in the process of being completed shall not be rejected due to identification of the defect provided the defect is considered not to cause a failure as defined below. If the defect will result in failures such as a split case, major primer leak, severe burn through, severe erosion, mouth tears, failure to extract or improper projectile extraction, any lots which contain the defect will be rejected. Parts processed after corrective action has been implemented shall be subject to special inspection until the contractor has demonstrated that the process is under control.

4.4.3.4 Anodic coating (Quality Assurance). Prior to qualification of the contractor's anodize process, the coating thickness and weight shall be determined for each lot in accordance with 4.5.3.5.1 and 4.5.3.5.2. Coating thickness and weight shall be determined in this manner on a minimum of three independent lots.

Once the anodize process has been qualified, lot acceptance testing shall consist of verification of film thickness on the outside diameter at the location (Region X of Figure 3) for AQL specified in 4.4.2.a, utilizing the non-destructive eddy current test method as described in 4.5.3.5.1.1. Methods of testing thickness shall be approved by the Government prior to implementation. Government approval will be given only after the selected method has demonstrated correlation with the thickness measurement using the microsection technique as defined in 4.5.3.5.1 for three independent anodized lots. See 6.5 for data submittal requirements.

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TABLE IV. Function and casualty test.

Case Casualty 1/	Ambient Acc	Ambient Rej	Low Temp. Acc	Low Temp. Rej	High Temp. Acc	High Temp. Rej
Failure to chamber 2/	0	1	0	1	0	1
Failure to extract 3/	0	1	0	1	0	1
Blown primer 4/	0	1	0	1	0	1
Split						
H or S	6	17	1	4	1	4
J	2	7	0	3	0	3
K, L or M	0	1	0	1	0	1
Rupture, partial 5/						
S or J	2	7	0	3	0	3
K or L	0	1	0	1	0	1
Rupture, circumferential separation	0	1	0	1	0	1
Detached metal						
Rim shear	0	1	0	1	0	1
Neck shear 6/	2	7	0	3	0	3

1/ For defect definitions, see AMCR 715-505, Volume 8. The case lot will not be penalized for the occurrence of malfunction(s) which are determined to have been caused by the test weapon or associated equipment.

2/ The case lot will not be penalized if malfunction cannot be shown to have resulted from a defective case.

3/ A fired sample cartridge case remaining in the chamber after the normal extraction cycle will be classed defective, unless it can be shown that the malfunction was caused by a defective weapon.

4/ The sample cartridge case shall be classed defective only if it is evident by visual inspection that both the primer pocket and case head are enlarged and deformed.

5/ Refer to Figure 5 for classifying splits and ruptures resulting from the function and casualty test. If a split or partial rupture extends into two or more defined areas, only the most severe defect criterion of Table IV for the areas involved shall be used.

6/ The case lot will not be penalized if malfunction cannot be shown to have resulted from a defective case. There must be evidence of detached metal from fired case exterior. The lot will not be penalized for the detachment of metal shavings from the projectile crimped area of the case interior.

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4.4.3.5 Unlisted firing defects. The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification, occurs in any firing test.

4.4.3.6 Material analysis. The contractor shall obtain certificates of conformance with the chemical analysis requirements of 3.3.2.2 from the material supplier for every heat lot of material used in the manufacture of the case. The contractor shall also conduct a chemical analysis, in accordance with the test methods of 4.5.3.8, on a random sample basis on each raw material shipment.

Each cartridge case lot shipment shall include copies of all applicable chemical analysis certificates obtained from the raw material suppliers. In addition, copies of the actual chemical analysis performed by the case supplier for the applicable raw material shipment shall be included with the shipment. See 6.5.

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4.5 Test methods and procedures.

4.5.1 Performance tests. The performance tests shall be conducted on a complete round which has been fabricated and assembled in accordance with the applicable specifications.

4.5.1.1 Case rupture. The round shall be fired in a test barrel which conforms to Drawing 11825969. The case shall be visually examined for evidence of case rupture.

4.5.1.1.1 Chamber over-pressure. Special test rounds shall be assembled in which the propellant load is such that the chamber pressure, measured at the case mouth, is within the range of 380 to 415 MPa. The round shall be fired in a test fixture barrel. The cartridge cases shall be examined for evidence of case rupture. The case shall be considered to have failed if it ruptures at a pressure of less than 415 PMa. Cases tested at pressure in excess of 380 MPa shall be considered a success if no failure occurs and will be considered a no-test if failure occurs at pressures above 415 MPa.

4.5.1.2 Case extraction. With the round fired in a test fixture barrel or equivalent, the case shall withstand the extraction forces without failure of the extractor ring.

4.5.1.3 Projectile extraction. The cartridge case shall be constrained in a test fixture utilizing a chamber simulating ring and an axial load applied to the projectile at a withdrawal rate of 25 $\pm$ 10% mm per minute. The force required to extract the projectile from the case shall conform to 3.2.1.3.

4.5.1.4 Automatic gun tests. The cartridge case shall not degrade automatic gun performance, or cause gun stoppage, and shall not rupture.

4.5.2 Environmental tests. Unless otherwise specified, performance during or after environmental exposure, as specified, in the applicable test procedure, shall consist of conducting the tests of 4.5.1.1. and 4.5.1.2. Unless otherwise specified, these tests shall be conducted on complete target practice cartridge loaded in accordance with 3.2.1.

4.5.2.1 High temperature. The cartridge shall be subjected to the high temperature test of MIL-STD-810, Method 501, Procedure I. The operating temperature shall be 71°C. With the cartridge temperature stabilized at 71°C, the test item shall immediately be subjected to the tests referenced in 4.5.2.

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4.5.2.2 Low temperature. The cartridge shall be subjected to the low temperature test of MIL-STD-810, Method 502, Procedure I. For this test the storage and operating temperatures shall be -54°C. With the cartridge temperature stabilized at -54°C, the test item shall immediately be subjected to the tests referenced in 4.5.2.

4.5.2.3 Salt fog. The cartridge shall be subjected to the salt fog test of MIL-STD-331, Method 107 for 48 hours. After exposure, the cartridge case shall be visually examined for evidence of corrosion and the cartridge subjected to the tests referenced in 4.3.2.

4.5.3 Visual and mechanical.

4.5.3.1 Profile. The case shall be inspected for compliance with the dimensions shown on Drawing 11825902.

4.5.3.2 Weight. The case shall be weighed. The weight shall be recorded for information only.

4.5.3.3 Volume. The cartridge case primer hole shall be sealed flush with the interior surface of the base of the case. The case shall then be filled with liquid flush with the top of the case and the volume of the liquid measured. The volume shall be recorded for information only.

4.5.3.4 Structural characteristics.

4.5.3.4.1 Material mechanical properties. Test specimens consisting of half-inch wide rings shall be taken from Regions B and C shown in Figure 1. Any anodize finish which is on these specimens shall be removed prior to flattening by immersion in the following solution:

Phosphoric acid, 85%	-	35 milliliters
Chromic acid (CrO_3)	-	20 grams
Water to make	-	1,000 milliliters

To speed film removal, this solution may be heated to 100°C.

The two rings shall be measured for thickness variation and then cut through the thickest area. Trim equal amounts from each end to any convenient length and flatten to form a single tensile coupon strip representing each Region (B and C). Any flattening technique is acceptable provided it does not produce scratches on the significant surface or work harden the material to the extent that the mechanical properties will be effected. Tension leveling the two - high mill rolling are not acceptable as they significantly effect the mechanical properties.

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Unless otherwise specified coupons shall be milled to a minimum reduced section width of 0.250 $^{+0.010}_{-0.002}$ inch. To take advantage of various tensile machine test ranges for yield strength determination the coupon width in the reduced section area may be somewhat wider provided prior approval is obtained from the contractor. The $^{+0.010}_{-0.002}$ inch width tolerance will hole for any such dimensional change approved.

Width variations over the length of the reduced section and location of gage length shall be as specified in ASTM E8, Figure 6 Rectangular Tension Test Specimens. The overall length of the reduced section shall be 1-1/8 inches minimum with a radius of fillet generous enough to preclude the possibility of tensile coupon failure outside the gauge marks. Width of the grip section shall be one half inch.

The thickness dimensions shall be measured with a double ball micrometer and the width dimension with a blade micrometer. All measurements shall be read and recorded to the nearest 0.0001 inch. Coupons will be tested for tensile strength, yield strength and elongation in accordance with ASTM E8. The load ranges selected for testing shall be approximately 500 pounds full scale for the mouth and 1000 pounds full scale for the sidewall. In no case shall the pounds per division on the chart paper exceed 10 for the sidewall and five for the mouth.

The data from all cases tested shall be combined and the average values determined. The average for each property must meet the specified requirement of Table I. In addition, individual readings shall meet the requirements of Table II. All mechanical property shall be reported on the certification format of Figure 4 or equivalent.

4.5.3.4.2 Hardness. The mouth, the sidewall and the head shall be hardness tested by taking three readings in each location. The test cases shall be the same samples as those used in 4.5.3.4.1. Specific areas for these tests are shown in Figures 1 and 2. Other locations may be tested for conformance should any question of compliance with the specified limits arise. The surface to be tested shall be prepared so it is free of any protective finish and roughness which may effect the hardness reading. hardness testing shall be done before the sample is flattened for the mechanical properties testing (4.5.3.4.1).

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Head hardness shall be taken near the half-radius on the test surface shown on Figure 2. The sample shall be prepared such that the surface being tested and its opposite surface are parallel. Adequate care shall be taken in cutting the section so that softening of the cut surface is avioded and smearing of the surface layers is kept to a minimum. Any residual smeared metal shall be removed by conventional metallographic wet grinding techniques.

Rockwell 30T and B test methods shall be consistent with testing machine manufacturer recommendations. Other technique and description of apparatus for testing shall be in accordance with Federal Standard No. 151, Methods ASTM E18 or ASTM E92.

Hardness testing must be conducted on the specified scales. Conversion from any scale to another is not allowed. Excluding the head area, Rockwell 30T values shall be taken on the inside diameter of the case.

Referring to Region A, if any of the individual values are less than the allowable minimum the lot shall be rejected. If hardness taken on tensile coupons in Regions B and C does not comply with the requirements of Tables I and II but all the mechanical properties are within their allowable ranges the lot may be accepted on the basis of mechanical property data at the discretion of the Government. All hardness test results shall be reported on the certification form of Figure 4 or equivalent.

4.5.3.4.3 Mouth hardness (in-process mouth anneal verification). The routine hardness test utilized during the mouth thermal treatment process (Ref. 3.3.2.5) may be defined by the supplier. The method utilized shall have demonstrated the capability to provide data which correlates with data obtained in 4.5.3.4.2. Cases subjected to a non-destructive test may be returned to the production process provided multiple readings are separated by 90° and no other is sustained which may cause mouth erosion, burrs, or splits.

4.5.3.5 Anodic coating. Prior to qualification of the case manufacturer or anodizing processor (Ref. 4.4.3.4) the coating weight and thickness by (microsection) will be done and compared to the requirements shown in Figure 3. After qualification, when the supplier has been notified by the contractor, the thickness will be measured by the approved NDT eddy current technique. The coating weight will still remain a requirement after qualification but will no longer be tested on a routine basis unless specifically requested.

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During the qualification phase those cases selected for thickness measurement by the microsection technique shall also be tested for thickness by a non-destructive eddy current test method. Measurements by both techniques shall be taken in the same location of the case to optimize repeatability. The correlation thickness data thus developed will be the basis for approval of the suppler's NDT thickness test method. The thickness values obtained by both methods shall be recorded on the certification form of Figure 4 or equivalent.

4.5.3.5.1 Thickness (microsection technique). The cartridge case shall be cut such that two rings, (shaded portion of Figure 3) can be extracted for coating weight and thickness measurement. A narrow vertical strip, 1/8 to 3/16 inch wide, shall be cut from each for thickness determination. Each strip shall be mounted by conventional metallographic technique. Thickness measurements shall be taken at a minimum of three places on the inside and outside of each strip. The data thus obtained shall be average for the inside and outside of each sectioned strip.

The four average thickness values for each case shall meet the requirements of Figure 3.

4.5.3.5.1.1 Thickness (NDT eddy current method). After qualification of the anodize process and source used by the case manufacturer the anodic film thickness will be tested on the outside diameter of Region X of figure 3 by taking one reading per sample. If this reading is satisfactory then the next case in the sample will be tested. A case with this single reading outside the required tolerance range will be retested in four locations 90° apart. These test results will be averaged and if the average value is within the range requirement the case will be accepted. If this average is outside the allowable range requirement the suspect case may be cross sectioned through the area found to be the farthest out of the range requirement. The procedure of 4.5.3.5.1 will be used to determine the average thickness. If this average thickness value is within the range requirement the case will be considered acceptable.

4.5.3.5.2 Weight. The remainder of each ring obtained by the procedure described in 4.5.3.5.1 shall be tested for coating weight per 4.6.1 of MIL-A-8625. The individual coating weights of all three parts shall conform to the requirements of Figure 3. The coating weight data thus obtained will be reported on the certification form shown in Figure 4 or equivalent.

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4.5.3.6 Workmanship. The case shall be examined for compliance with the workmanship requirements of 3.5.3 and the applicable drawing. Where defects cannot be clearly defined by written descriptions, visual standards to define limits of acceptability shall be established by mutual agreement between the supplier and contractor. Workmanship evaluation for the preproduction tests shall include cutting the cartridge cases to permit examination for both internal and external characteristics.

4.5.3.7 Non-destructive tests. The cartridge case shall be subjected to the non-destructive tests (NDT) defined below to verify case integrity against selected defects standards. The use of contractor or supplier furnished NDT equipment and procedures shall be defined in the purchase order and shall be approved by the contractor prior to implementation. Defect standard (see 6.7), approved by the contractor, shall be used for calibration of the NDT equipment prior to and peridically during inspection testing.

4.5.3.7.1 External defects, case sidewall. Examination of the case sidewall for longitudinal cracks or seams shall be performed using either a binocular microscope using a minimum of 7X magnification or a (NDT) device capable of detecting external cracks or seams. A 200% inspection shall be employed when using the optical method. This shall be accomplished in a tandem arrangement with the second operator examining the case immediately following the first operators examination. Prior to the microscopic examination the entire surface of each case will be examined for random gross defects. When using the optical method, adequate lighting must be available and higher magnification can be used to differentiate between a scratch or a crack. When it is still questionable whether or not a crack exists, then die penetrant testing per MIL-I-6866 can be employed. Any case found to have a crack in excess of 38.1 mm long by .19 mm deep shall be considered as a defect. The cross sectioning technique may be used to verify the depth of a crack identified visually. When the microscopic technique is employed the area of inspection will extent from 15.24 mm to 85 mm from the head end of the case.

When a (NDT) method is used, sufficient standards will be available to verify that cracks or seams in excess of .19 mm deep by 6 mm long can be reliably identified. Any cracks or seams in excess of .19 mm deep by 6 mm long shall be considered as defects. The area of inspection to be included in the NDT technique shall extend from 15.25 mm to 85.0 mm when measured from the head end of the case.

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4.5.3.7.2 External defects, case mouth. This inspection shall be performed using either a binocular microscope using a minimum of 7X magnification or a (NDT) device capable of detecting external cracks or seams in the first 7 millimeters (mm) of the case mouth. When using the optical method adequate lighting must be available and higher magnification can be used to differentiate between a scratch or a crack. When it is still questionable whether or not a crack exists, then die penetrant testing per MIL-I-6866 can be employed. Any case found to have a crack in excess of 0.25 mm deep shall be considered as a defect. The cross sectioning technique will be used to verify the depth of a crack identified visually. Refer to 6.7.

When a (NDT) method is used, sufficient standards will be available to verify that cracks or seams in excess of 0.25mm deep by 1.27mm long can be reliably identified. Any cracks or seams in excess of 0.25mm deep shall be considered as defects.

4.5.3.7.3 Internal defects, sidewall. This inspection shall be performed using a visual technique or an (NDT) technique capable of detecting internal sidewall defects having minimum dimensions of .25 mm deep by 8.9 mm long. This examination applies to an area measured between 13.0 mm and 28.0 mm from the external base of the case.

The visual technique shall use a light source to highlight any internal sidewall discontinuities. A crack will appear as a thin, highly reflective line with usually an aluminum color. Its orientation will be circumferential in most cases. Any suspect crack cases will be verified by the die penetrant technique of MIL-I-6866. Any case found to have a crack in excess of .19 mm deep shall be considered as a defect. The cross sectioning technique will be used to verify the crack depth. For definition, refer to 6.3.7.

When the)NDT) method is employed sufficient standards will be available to verify that cracks or seams in excess of .25 mm deep by 8.9 mm long can be reliably identified. Cracks or seams in excess of this depth and length shall be considered as defects.

4.5.3.8 Chemical analysis. A sample quantity of the case material shall be subjected to a chemical composition test. The individual values thus obtained shall be included on the certification form of Figure 4 or equivalent.

4.5.3.9 Case wall thickness. The case shall be cross sectioned and wall thickness measured to verify conformance with the dimensions of the applicable drawing.

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4.5.3.10 Special defects tests. The cartridge case shall be subjected to the non-destructive tests defined below to verify case integrity or characteristics against selected defect standards. The use of contractor or supplier furnished test equipment and procedures shall be defined in the purchase order and shall be approved by the contractor prior to implementation. Defect standards (Ref. 6.3.8), approved by the contractor, shall be used for calibration of the test equipment prior to and periodically during inspection testing.

4.5.3.10.1 Presence of mouth anneal. This inspection shall be done using either a device for measuring electrical conductivity or by hardness test.

The electrical conductivity test technique shall have a sensitivity to detect a part which has never been mouth annealed from one which has been mouth annealed. Standards exhibiting these various conditions will be used to calibrate the device.

If hardness testing is used one reading shall be taken on the outside diameter near the center of Region C in Figure 1 to determine the acceptability of the part. If a no-test condition is encountered a second reading may be taken but it must be located 90° away from the first reading if the part is to be returned to the lot.

During this test, the cases which fall outside the typical range for a mouth annealed part will be set aside in two groups one for being too low and one for being too high in hardness. From each group set aside, up to three with the lowest hardness and the three highest hardness will be mouth mechanical property tested per 4.5.3.4.1. The test results must meet the mechanical property requirements of Table II or the lot will be rejected.

The hardness shall comply with the requirements of Table II when corrected for anodize finish effects and radius of curvature effects. The correction factor shall be determined on a minimum of six cases initially rejected, by comparing the initial outside diameter hardness with the hardness obtained using the technique of 4.5.3.4.2. Documentation of all test data generated to determine any correction factor will be retained for review by the procuring activity upon request.

4.5.3.10.2 Completeness of heat-treatment. The inspection shall be performed using either an eddy current type device capable of measuring change in electrical conductivity of aluminum or by hardness test.

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The eddy current test shall consist of placing the instrument probe against the head end of the case and comparing the conductivity value to those of the standards used to calibrate the equipment. The sensitivity of the equipment shall be such that is can detect product which is properly heat treated from product which has missed one or more of its hardening thermal treatments. Standards shall be used to verify this fact.

All standards will be stored for 100 hours at room temperature before being used. The actual conductivity value obtained on each standard shall be recorded and monitored to assure that the standard continues to be acceptable.

If hardness testing is employed, it shall be conducted using the Rockwell B scale. Tests shall be taken in Region A of Figure 1. The part shall be properly supported and shall be tested per the applicable sections of Federal Standard No. 151, Method ASTM E18. Standards which have missed one or more of the hardening thermal treatments shall be used to calibrate the test setup. Hardness tests may be conducted on the head surface at one half radius from the primer pocket to extractor rim. A 2.5 point RB correction factor must be applied to this method, i.e., reject below RB 87.5. Hardness tests may also be conducted on the side of the head at the bottom of the extractor groove. A 1.0 point RB correction factor must be applied to this method, i.e., reject below RB 86.0.

Any cases found to be less than the minimum hardness value, using the half radius head surface test, may be retested using the extractor groove method and accepted on the basis of the retest.

4.5.3.11 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.11 herein.

5. PACKAGING

5.1 Packaging and packing. Cartridge cases shall be preserved, packaged and packed in accordance with the contractor best commercial practice to insure acceptance by common carrier for safe transporation, at the lowest rate, to the point of delivery. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with the latest revision of MIL-STD-129.

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6. NOTES

6.1 Intended use. The component covered by this specification is intended for use in M788, M789 and M799 series of 30MM Ammunition.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment design as required to Commander, ATTN: DRDAR-QAF-I, ARRADCOM, Dover, NJ 07801.

6.4 Visual defects standards. Items which have been rejected during production shall be segregated and utilized for the visual standards established jointly by representatives of the supplier and contractor. The standards selected shall include both internal and external defects. Subsequent application of these standards to acceptance inspecion shall be specified in the contract or purchase order.

6.5 Data report. All data obtained from testing as described in 4.2.2.2, 4.2.2.4 and 4.2.2.5 shall be provided on the cerification format of Figure 4 of equivalent and shall accompany each cartridge case lot.

6.6 Case mouth annealing process. The supplier shall submit with each confirmed purchase order a description of the thermal testing process to be used to obtain the desired mouth mechanical properties. This document should contain but need not be limited to the following information.

Type of heating medimum _____.

Temperture uniformity of heating medimum _____.

Immersion depth range while in heating medimum _____.

Minimum time delay before rinsing _____.

6.7 Non-destructive test standards. The contractor and supplier shall establish defect standards for use in the NDT tests defined in 4.3.3.7.1 and 4.3.3.7.3. The standards shall be approved by the Government.

6.8 Special defects standards. The contractor and supplier shall establish defect standards for special defect tests defined in 4.3.3.10.1 and 4.3.3.10.2. The standards shall be approved by the contractor.

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6.9 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.10 First article sample. In order to minimize inspection costs, the firing tests of 4.3.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cases may be required by the testing facility.

6.11 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. The letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

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